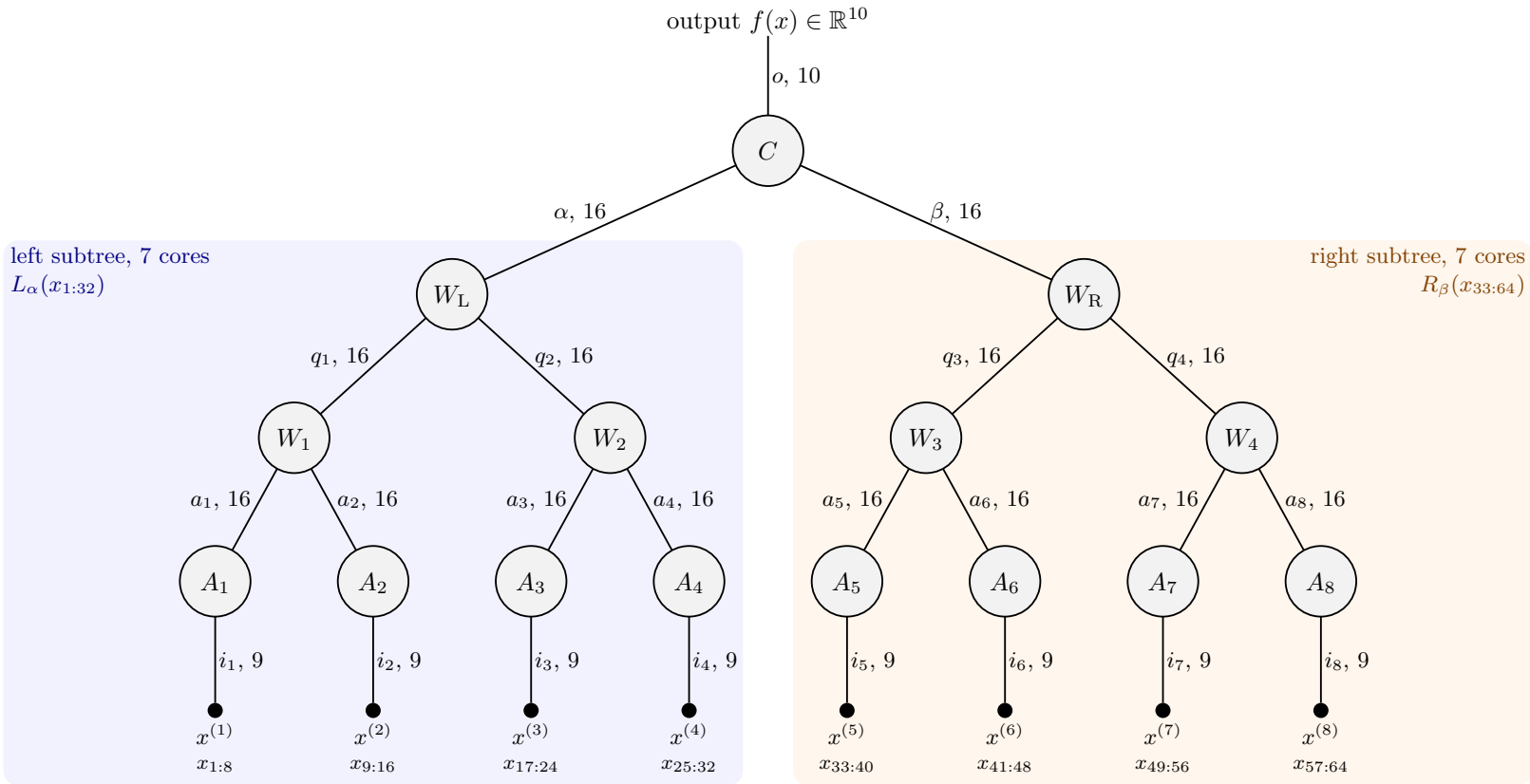


$C_{o\alpha\beta} \in \mathbb{R}^{10\times16\times16}$
2,560 params

$W_{\mathrm{L}}, W_{\mathrm{R}} \in \mathbb{R}^{16\times16\times16}$
$2 \times 4{,}096$ params

$W_1, \ldots, W_4 \in \mathbb{R}^{16\times16\times16}$
$4 \times 4{,}096$ params

$A_1, \ldots, A_8 \in \mathbb{R}^{16\times9}$
8×144 params

$\phi(x^{(j)}) = (1, x^{(j)}) \in \mathbb{R}^9$
data, not trained

total trainable = 28,288

$$f_o(x) = \sum C_{o\alpha\beta}\,(W_{\mathrm{L}})_{\alpha q_1 q_2}\,(W_{\mathrm{R}})_{\beta q_3 q_4} \prod_{k=1}^{4} (W_k)_{q_k a_{2k-1} a_{2k}} \prod_{j=1}^{8} (A_j)_{a_j i_j}\,\phi(x^{(j)})_{i_j}$$

sum over all bond indices $\alpha, \beta, q_k, a_j, i_j$; fusion-core indices ordered (parent, left child, right child); bonds labeled "index, dimension"